Diagnose, repair, recheck

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify a reasoning error and demonstrate the repaired method on a new problem.

SUCCESS CRITERIA - I can...

1. Locate the first incorrect step.
2. Show a corrected method with units.
3. Apply the repair to a different problem.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “evidence” and “revision” to explain your answer. Sentence starters are optional.

evidence: observations or measurements that support an idea

revision: a change made after checking evidence or reasoning

assumption: a condition accepted for using a model

01 Warm-up: bring back an earlier idea

A **3 kg** cart moves left at **5 m/s**. Forces are **26 N** right and **5 N** left. **Find acceleration** and whether its speed initially rises or falls.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $a = \Sigma F/m$; $v_f = v_i + a\,\Delta t$.
For a constant-acceleration interval. **Check signs**, units and what the equation treats as true; keep the original step visible when revising.

Worked reasoning: A student says that a car moving *east* must accelerate *east*. A car that moves *east* while braking is one example that shows this is wrong: its acceleration points *west*. One example can disprove a statement that says something is always true.

Your turn: 1. **Draw velocity** and acceleration arrows for that counterexample.
2. **Write a corrected statement about** their directions.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Car braking while moving east

East is right.

Velocity:

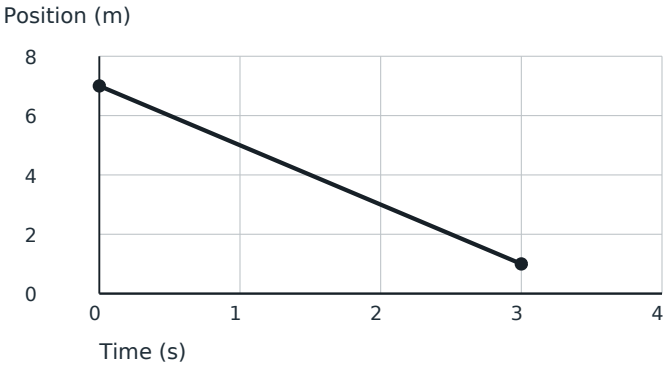
Acceleration / force:

Marzano check - circle: ① ② ③

03 Position is not velocity

A cart's position changes from **7 m** to **1 m** in **3 s**, as shown. A student says, "Its velocity is **1 m/s** because it ends at **1 m**." Explain the mistake and **calculate the velocity correctly**.

Sentence starter: The step I would change is ... . I would change it because I found ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Net force is not one force

A **4 kg** cart has **18 N** right and **6 N** left. A student calculates acceleration as **18 ÷ 4**. **Use the force** diagram to **explain the missing step**, then correct the calculation.

Sentence starter: I think … . I can point to …, which shows …, so … .

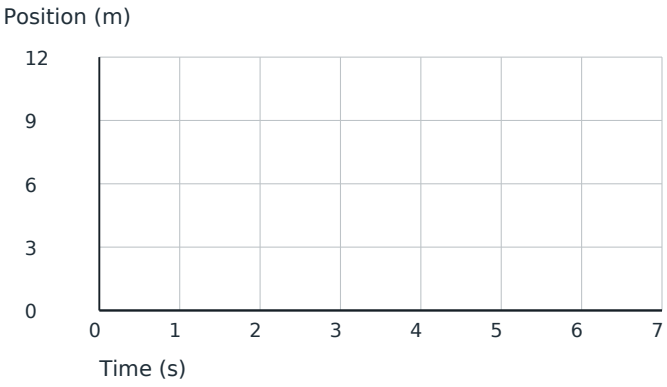


Marzano check - circle: ① ② ③

05 New measurements or observations

Try two new checks.
(a) A robot moves steadily from **10 m** to **2 m** in **7 s**. **Find its velocity**.
(b) A **5 kg** cart has **7 N** right and **22 N** left. **Find its acceleration**. **Show how each calculation accounts** for direction.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A cart travels *west* at **14 m/s** and *west* at **5 m/s 4 s** later. Is it speeding up? **Find acceleration size** and direction. **Sketch the velocity**–time graph; **label the given values**.

Word help: acceleration = rate of change of velocity

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

07 Exit check

1. **Choose the error** in question **3** or **4**.
2. **Explain the correction**, then point to the matching new check in question **5**.
3. What can you now do without help?

Sentence starter: The step I would change is … . I would change it because I found … .

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Invent a wrong statement about force or motion, then **draw a specific case** that disproves it. **State the corrected rule**.

Sentence starter: The step I would change is … . I would change it because I found … .

Add and label force arrows.

Use a diagram to test the statement you invent.

Marzano check - circle: ① ② ③